

EDUCATION

- 2018.3 – 2023.2 **Ph.D. in Applied Bio-Engineering**, Seoul National University
Thesis: *Clinical Information Extraction from Unstructured Free-texts for Pharmacovigilance* · Advisor: Howard Lee
- 2013.3 – 2018.2 **B.S. in Biology and Anthropology**, Seoul National University — Summa Cum Laude

EXPERIENCE

- 2026.1 – Present **Sr. AI Engineer**, Selta Square
- 2024.8 – 2025.12 **Research Assistant Professor**, Biomedical Research Institute, Seoul National University Hospital
- 2023.3 – 2024.7 **Post-doctoral Researcher**, Seoul National University Hospital

RESEARCH INTERESTS

Clinical / Document Information Extraction · Biomedical Natural Language Processing · ML-based Drug Development · Clinical Informatics · Clinical Pharmacology & Regulatory Science

PUBLICATIONS

Journal Articles

- Intra-session recurrence in intradialytic hypotension prediction: evaluation implications and recurrence-aware modeling.**
Siun Kim, Jiwon Ryu, Sejoong Kim, Su Hwan Kim, Myeongju Kim, Hyung-Jin Yoon. *Frontiers in Digital Health*, 2026.
- Beyond fine-tuning: leveraging domain-aware in-context learning with large language models for clinical named entity recognition.**
Siun Kim, David Seung U Lee, Yujin Kim, Hyung-Jin Yoon, Howard Lee. *Journal of Biomedical Informatics*, 2026.
- One-year post-acute COVID-19 syndrome and mortality in South Korea: a nationwide matched cohort study using claims data.**
Jung-Hyun Won, Yesol Hong, Siun Kim, Howard Lee. *Frontiers in Public Health*, 2024.
- Deep learning-based long-term risk evaluation of incident type 2 diabetes using electrocardiogram in a non-diabetic population: a retrospective, multicentre study.**
Junmo Kim, Hyun-Lim Yang, Su Hwan Kim, Siun Kim, Jisoo Lee, Jiwon Ryu, Kwangsoo Kim, Zio Kim, Gun Ahn, Doyun Kwon, Hyung-Jin Yoon. *eClinicalMedicine*, 2024.
- Automatic extraction of comprehensive drug safety information from adverse drug event narratives in the Korea Adverse Event Reporting System using natural language processing techniques.**
Siun Kim, Taegwan Kang, Tae Kyu Chung, Yoona Choi, YeSol Hong, Kyomin Jung, Howard Lee. *Drug Safety*, 2023.
- An annotated corpus from biomedical articles to construct a drug-food interaction database.**
Siun Kim, Yoona Choi, Jung-Hyun Won, Jung Mi Oh, Howard Lee. *Journal of Biomedical Informatics*, 2022.
- A population pharmacokinetic-pharmacodynamic model of YH12852, a highly selective 5-hydroxytryptamine 4 receptor agonist, in healthy subjects and patients with functional constipation.**
Siun Kim, Hyun A Lee, Seong Bok Jang, Howard Lee. *CPT: Pharmacometrics & Systems Pharmacology*, 2021.
- A critical review of the United States regulatory pathways for determining the equivalence of efficacy between CT-P13 and original infliximab (Remicade).**
Soohyun Kim, Siun Kim, Howard Lee. *Drug Design, Development and Therapy*, 2020.
- Improving clinical data standardization in surgical notes through biomedical entity linking with contextual evidence from large language models.**
Chaiho Shin, Dareen Eom, Hahyun You, Kwangsoo Kim, Siun Kim*, Hyung-Jin Yoon*. (*co-corresponding) *BMC Medical Informatics and Decision Making*, 2026.

Conference & Workshop Papers

10. **DiZiNER: disagreement-guided instruction refinement via simulating pilot annotation for zero-shot named entity recognition.**
Siun Kim, Hyung-Jin Yoon, Howard Lee. *Proceedings of ACL 2026 (Main Conference)*, 2026.
11. **Questioning our questions: how well do medical QA benchmarks evaluate clinical capabilities of language models?**
Siun Kim, Hyung-Jin Yoon. *24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025) — Best Paper*, 2025.
12. **Large language model-assisted systematic review: validation based on Cochrane review data.**
Siun Kim, Hyung-Jin Yoon. *Medical Informatics Europe (MIE) 2025*, 2025.
13. **CRese: benchmark data and automatic evaluation framework for recommending eligibility criteria from clinical trial information.**
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Yingce Xia, Tao Qin, Howard Lee. *Findings of EACL 2024*, 2024.
14. **Revisiting machine learning based drug repurposing: drug indications are not a right prediction target.**
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Yingce Xia, Tao Qin, Howard Lee. *Conference on Health, Inference, and Learning (CHIL 2023)*, PMLR, 2023.

Under Review

Book

15. **The era of biologics is coming.**
Siun Kim, Howard Lee. (Selected as 2022 Sejong Book) *Young Doc Press*, 2019.

PROJECTS

Seoul National University Hospital

How well medical QA benchmarks reflect the clinical capabilities of LLMs 2025.2 – 2025.8

Systematically correlated LLM scores on major medical QA benchmarks (MedQA, MedMCQA, PubMedQA, MMLU medical subjects) with real-world clinical performance; benchmarks only partially reflect clinical capability. (Spearman $\rho = 0.59$ (moderate))

Zero-shot NER via disagreement signals across multiple LLMs 2023.3 – 2025.10

Model-iteration configuration selection outperforming instruction fine-tuned baselines.

Key patient-information extraction from Hematology-Oncology clinical notes 2023.3 – 2025.10

Extracting diagnoses, symptoms, and prescriptions using few-shot retrieval and LLM in-context learning.

Intra-dialysis hypotension (IDH) prediction from TDMS time-series vital-sign and dialyzer data SNUH · 2023.3 – 2025.8 · Lead researcher

Recurrence-aware modeling and evaluation for long-term dialysis patients. (AUROC up to +8 pp)

Seoul National University

Revisiting ML-based drug repositioning with clinical-trial eligibility criteria Microsoft Research Asia · 2022.5 – 2023.4 · Project lead

Reformulation of ML-based drug repositioning with eligibility-criteria recommendation.

NLP algorithms for extracting key clinical information from drug adverse-event reports Ministry of Food and Drug Safety · 2021.2 – 2022.2 · Project lead

KAERS-BERT for mixed Korean-English narratives; ICH E2B(R3)-based annotated corpus; NER, RE, sentence extraction and label classification models.

Prospective real-world registry for pediatric ALL from multi-institutional EHRs Korea Health Industry Development Institute · 2022.3 – 2022.12

Report-form design and extraction-algorithm development.

Target Product Profile (TPP) of REX101, a bacterial-exosome anticancer therapy Rosetta Exosome · 2021.3 – 2021.12 · Lead researcher

IDX-1197 development strategy consultation Idience Co., Ltd · 2020.10 – 2021.2 · Lead researcher

Pre-clinical and clinical database construction for PARP inhibitors.

Drug-interaction evaluation using big data and machine learning Ministry of Food and Drug Safety · 2018.2 – 2022.2 ·

Subpart lead

NLP for drug-food interaction extraction and drug-drug interaction prediction.

PK/PD data analysis for Phase 1 trials of YH12852 and CWP232291 2018.3 – 2022.12

Analysis resulting in a population PK/PD model publication.

CONFERENCE PRESENTATIONS

CRese: benchmark data and automatic evaluation framework for recommending eligibility criteria from clinical trial information — *EACL 2024, Malta*, Mar. 2024 · Poster

How to address data scarcity and bias for AI-powered drug discovery — *2023 Fall Korean Society of Medical Informatics Conference, Seoul*, Nov. 2023 · Oral

An annotated corpus and NLP model for drug safety information extraction from free-text adverse-event reports — *2021 Fall Korean Society of Medical Informatics Conference, Seoul*, Nov. 2021 · Oral

Building natural language training data for clinical information extraction — *2021 Korean Society of Medical Informatics Conference, Yongin*, Nov. 2021 · Tutorial

A population PK-PD model of YH12852, a highly selective 5-HT4 receptor agonist — *Population Approach Group in Korea, Iksan*, Nov. 2020 · Oral

Development of an annotation guideline to extract drug-food interaction information for machine learning — *Korean Society for Clinical Pharmacology and Therapeutics, Yeosu*, Nov. 2019 · Poster

A population PK-PD model to characterize the increase in gastrointestinal motility by YH12852 — *2019 AAPS PharmSci 360, San Antonio, USA*, Nov. 2019 · Poster

AWARDS & HONORS

First Prize, Best Paper — The 24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025), Vienna, Austria, Aug. 2025

Best Article Presentation (1st) — Korean Society for Clinical Pharmacology and Therapeutics, Nov. 2021

Best Original Research Article (1st) — The Korean Society of Medical Informatics, Oct. 2021

Presidential Science Scholarship — Korea Student Aid Foundation, 2013 – 2017

PATENT

Apparatus and method for extracting drug-food interaction information — Howard Lee, **Siun Kim**, KR 10-2021-0172058 (application)

REVIEW EXPERIENCE

Conferences: ML4H (2024, 2025, 2026), MLHC (2025, 2026), BioNLP Workshop (2025)

Journal: JAMIA (2024, 2025, 2026)

SKILLS & LANGUAGES

LLM & Agents: OpenAI API, Anthropic API, Gemini API, Ollama, OpenRouter, LangChain

NLP & Retrieval: Hugging Face Transformers, Sentence Transformers, FAISS, BM25

ML & Statistics: PyTorch, scikit-learn, JAX, NumPyro, statsmodels, SciPy

Data: Python, pandas, NumPy, SQL, PostgreSQL, R

Pharmacometrics: NONMEM

Languages: Korean (native), English (business fluent)